

Keeping Data Safe and Secure on the Cloud



Cloud enables users to dynamically avail resources based on their needs, and only pay for what they use. That's a great invention for the digital world. But how secure is it? We take a look at the key elements involved in cloud security.

Cloud security is the shared responsibility of the customer and the cloud service provider (CSP). Though users technically own the servers, they generally don't have physical control over them. Authentication, authorisation, firewalls, audit and transaction visibility are crucial steps in cloud security.

Regions and availability zones

Uptime and availability are important aspects of cloud security. Service providers maintain their infrastructure in different geographic locations. For each region there can be more than one availability zone (AZ). For example, a cloud service provider could support two regions -- India and

Australia, and in each of the regions there could be multiple AZs. For instance, there could be two AZs in Mumbai in India but in different parts of the city.

Each of the AZs is self-contained and independent, having different power providers, separate network confirmations, and so on. This makes them reliable, as downtime in one AZ will not affect the other AZ.

As users we can take advantage of this, and configure our application to the nearest region in order to reduce latency. We can also load balance either through active/active or active/standby configurations. This provides the flexibility to switch when there is an issue or overload requests, enabling high availability.